

Trustworthy Causal Explanations: Uncertainty for Causal Dependence Plots

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Prerequisites:

- Python programming experience; comfort with numpy, pandas, and basic scikit-learn
- Probability and statistics: expectations, conditional distributions, regression, confidence intervals (we will cover causal-inference specifics)
- Interest in explainable AI and in whether its outputs can be trusted (important!)
- Bonus skills: any exposure to causal graphs, doubly robust estimation, the bootstrap, or designing simulation studies

Background and Motivation

Machine-learning models are often explained with *feature-effect curves*: plots of how a model's prediction changes as one input feature varies. The standard tool, the partial dependence plot, moves one feature *while freezing all the others*. This is not what would happen if that feature really changed in the real world, where it may causally influence other features. [Causal dependence plots] (<https://arxiv.org/abs/2303.04209>) (CDPs, NeurIPS 2024) address this by *propagating the change through a causal graph of the features*. They are meant for high-stakes uses: model audits, fairness assessments, scientific reporting.

But CDPs are currently drawn *without any statement of uncertainty*. The curve is estimated from a finite sample; the causal graph is a modelling assumption; an unmeasured factor could confound the plotted feature. An explanation that looks decisive but is statistically fragile manufactures unearned trust. *Responsible AI needs explanation methods that report their own uncertainty and their own assumptions.*

This project builds that on the foundation of a mature statistical toolbox (doubly robust estimation, cross-fitting, bootstrap confidence bands, sensitivity analysis). You will work in simulation environments where the true curve is known exactly, so that identities can be checked to machine precision and confidence bands can be checked for coverage with honest Monte-Carlo error bars. The project already has a verified, reproducible code base of such checks that you will extend.

Project Deliverables

1. A documented, open-source Python implementation of reduced-form estimators for the aggregate CDP family (total, direct, indirect, and binned/smoothed curves).
2. Simulation experiments with known ground truth and real-data case studies.
3. A project report and write-up targeting a conference submission (ICML, UAI, NeurIPS, or similar) with students as co-authors.

Project timeline

The following is a tentative timeline for Fall 2026 (13 weeks; dates per the program calendar). Supervision meetings at least weekly, other communication on Slack.

Weeks 1–2: Foundations

- Causal graphs, interventions, back-door adjustment on small examples; the partial dependence plot and the CDP family
- Run the project's verified check scripts; reproduce the key identity (total curve $\neq$ adjustment for the parents) by exact enumeration in toy models; learn the check-first workflow (seeds, tolerances, negative controls)

Weeks 3–5: The estimator at discrete feature levels

- Implement the doubly robust (AIPW) estimator with cross-fitting
- Compare adjustment sets: the parents of the feature versus all its non-descendants
- Coverage study on the simulation testbed with Monte-Carlo-aware tolerances

Weeks 6–7: Practical rules and the interaction plot

- Held-out explanatory data; the reference population; the overlap diagnostic
- Implement the "interaction dependence plot": the gap between the total curve and the sum of the direct and indirect curves

Weeks 8–9: Robustness to unmeasured confounding

- Planted confounders of known strength and sensitivity analysis
- Demonstrations of when adjustment and propagation agree and when propagation breaks

Week 10: Fall break / buffer

Weeks 11–12: Continuous features and the comparison study

- Binned levels and the smoothed curve with bootstrap confidence bands
- Propagation arm via DoWhy; head-to-head simulation; real-data examples; package documentation

Week 13: Wrap-up

- Finalize project report; paper draft
- Prepare presentation

With three students, weeks 8-12 split into parallel tracks (estimation and bands; robustness; comparison and case study); with two students, the tracks are merged and the case study is reduced in scope.